THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT
ANNUAL JOINT EXAMINATION

FORM THREE
033/BIOLOGY 1
MARKING SCHEME NOVEMBER 2024

SECTION A (16 Marks)

1.

(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)	(x)
B	B	A	E	C	D	B	C	D	A

2. ***01@ = 10 marks***

LIST A	(i)	(ii)	(iii)	(iv)	(v)	(vi)
LIST B	C	A	D	E	I	F

01@ = 06 marks

SECTION B (60 Marks)

3. (a)Burns and scalds

A burn is caused by dry heat; example, flame of hot object. A scald is caused by a steam or boiling water. The treatment is the same for both burns and scalds except that in scalds any wet clothing should be carefully removed while in burns clothing should be left in place. If clothes are burning cover the victim with a heavy blanket to cut off the air supply. (03 marks)

(b) First aid to a person with burns

- Dip the burning limb into cool water.
- Do not break the blisters.
- Cover the injured area with clean dry cloth to keep away bacteria entering the damaged tissue.
- If the situation is serious take the victim to the hospital immediately.

(Any three 01 @ = 03 marks)

First aid for chemical burns (acids and alkalis)

- Strip off all clothing that has been contaminated by the chemical.
- Wash the affected area with plenty of water.
- Apply moist packs soaked in a weak solution of baking soda.
- Vinegar can be used for alkaline burns.

(Any three 01 @ = 03 marks)

4. (a)

Nutrition is the process whereby organisms take in or manufacture nutrients and use them. Nutrients are substances that provide the body with energy and material for growth and repair essential for maintenance of life. *03 marks*

(b)

Heterotrophic nutrition is the nutrition whereby organisms use readymade food substances (complex organic molecules made by the autotrophs). Heterotrophic nutrition can be divided into holozoic, saprophytic and parasitic modes of nutrition. *01½ marks*

In Holozoic nutrition organisms ingest food and digest it to get the useful nutrients. Examples of holozoic organisms are human beings, birds and snakes. *01½ marks*

In Saprophytic nutrition organisms feed on dead and decaying organic matter by secreting enzymes and digesting the food externally and then absorbing the nutrients. An example of a saprophytic organism is the mushroom. *01½ marks*

In Parasitic nutrition organisms feed on digested food in another organism and harming that organism. An example of a parasitic organism is the tape worm *01½ marks*

5. Factors affecting rate of breathing
   - The concentration of carbon dioxide in the blood: breathing becomes faster and deeper because …
   - The concentration of haemoglobin: low concentration leads to faster breathing while high concentration lead to normal or slow breathing because…
   - Health status: rate of breathing increases with fever and other medical conditions
   - Altitude: rate of breathing increases at higher altitude because of the low levels of oxygen in the environment.
   - Physical activities: Rate of breathing with increased physical activities because of the increased expenditure of energy (01½@ = 09 marks)
   - Age: Rate of breathing decreases with aging as energy requirement and elasticity of structures decline

6.

(a) Skeletal muscles are voluntary…Cardiac muscles are involuntary
Skeletal muscles are fatigued….cardiac muscles are not fatigued 01½ @ = 04½ marks
Skeletal muscles are connected to the bones…cardiac muscles are located in the heart.

(b) Movement: the skeleton facilitate movement in cooperation with muscles….
Protection: the skeleton protects internal organs…
Support: the skeleton support the body from collapsing…01½ @ = 04½ marks

7. **(**a) Adjusting Over-heating through sweating in humans and panting in dogs:

Similarity: in both, Evaporation of moisture brings about a cooling effect. In both cases moisture is lost.

Differences; Sweating occurs all over the humans body, which their main method of losing heat. While animals of the dog family have sweat glands only in the pads of their paws. Their main method of losing heat is to pant rapidly with the tongue hanging outside. *(03 marks)*

(b)**Mechanism of Osmoregulation**

If the blood is too concentrated, the hypothalamus stimulates the pituitary gland to secrete into the blood the **ADH** (anti diuretic hormone) which causes the kidney tubules to absorb more water from glomerular filtrate back into the blood. Therefore, the urine becomes more concentrated and the blood more diluted.

If the blood is too diluted, production of **ADH** is suppressed and less water is absorbed from glomerular filtrate, thus urine becomes more diluted as a means of eliminating excess water from the body. *(03 marks)*

Mechanism of blood sugar regulation

When **blood sugar level increases**, the **beta cells** in the pancreas will produce **Insulin**, which stimulates the liver cells to extract more glucose from the blood and also promotes uptake of glucose by muscle cells. Liver **converts glucose into glycogen** which is be stored in the liver.

When there is **less glucose in the blood** the pancreas slows down the production of insulin and produces **glucagon** from the alpha cells, which converts the stored **glycogen into glucose** which is then released into the blood. When all glycogen has been used up, then stored fat is converted into glucose. The failure of the pancreas to produce sufficient insulin leads to **Diabetes**. The diabetic cannot *(03 marks)*

8.

(a) Components of nervous coordination

The components of the nervous system are stimulus, receptors, coordinator, effectors and response.

Stimulus: Stimulus is the change in the internal or external environment to which an organism responds. Types of stimuli: (1) physical (2) chemical (3) biological.

Receptors: Receptors are specialized cells that detect (receive) stimulus. The receptors (sensory cells) are located in sense organs. Examples of **sense organs** are: ***Skin*** (touch, pain, temp, & pressure, ***Ear (***sound and body position, balance), ***Eye*** (sight) and ***Tongue*** (taste).

Coordinator: This is made up of the neurones (nerve cells). 01 @ = 05 marks

Effectors: These are the organelles, organs and other structures that enables the organisms to respond to the stimulus.

Response: This is the reaction to a stimulus.

(b)

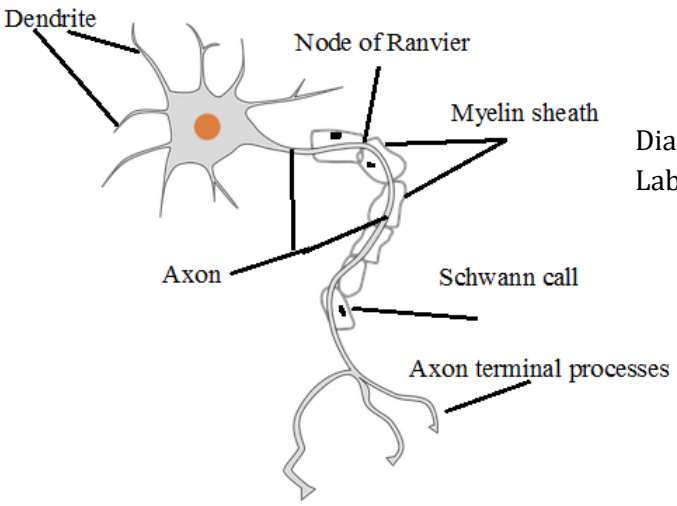


Diagram 2 =02 marks
Label: 4 × ½ = 02 marks

SECTION C (30 Marks)

9. Introduction:. Drug abuse is the indiscriminate and excessive use of psychoactive drugs for non-medical reasons with no regard to their side effects. While Drug addiction is the state of dependence caused by regular use of psychoactive drugs (or, compulsive desire to use drugs despite their harmful effect). *(1½ marks)*

Types of Drug Addiction

- Psychological or mental addiction: person thinks that his/her well being depends on the drugs and is unable to stop taking it, i.e. user has an emotional need. *(@1½ = 03 marks)*
- Physical addiction: the body needs the drug in order to work properly. If there is no drug the person suffers withdrawal symptoms.

Effects of Drug Abuse

- Cigarette smoking can lead to lung cancer and heart diseases.
- Alcohol can cause brain damage.
- Some drugs cause ulcers and rotten teeth.
- Many drugs damage the brain and other organs.
- Many drugs lead to addiction.
- Some drugs may lead to kidney damage.
- Pregnant women can affect their unborn baby.
- Drug addicts fail to concentrate at work (poor outcome).
- Drugs weaken our bodies- poor development of the society.
- Addicts indulge in irresponsible behaviour which can lead to unwanted pregnancies, STD's and HIV/AIDS.
- Student addicts mostly drop out of school or absent themselves. *(Any six @1½ = 09 marks)*
- May lead to traffic offences and accidents.
- Economic burden to others as the addicts do not work.
- Drug addiction can lead to marriage breakdown.
- Drug addiction lead to more crime.

Measures to Control Drug Abuse

- Avoid taking any drug without doctor's prescription.
- Keep off from peer pressure and addicted people.
- Be principled and say no to anyone who offers the drug to you.
- Use your leisure time properly (for sports and hobbies).
- Inform community leader or police of any use or trafficking of drugs.
- Form counselling club to educate people on how to keep off from drugs (education on proper use of drugs)
- Addicts should seek help from health officials.

Conclusion: Drugs are used as medicine, to alleviate pains and fight diseases. However, the indiscriminate and excessive use of psychoactive drugs for non-medical reasons creates a problem. So while we cannot eliminate the use of drugs, we should make sure they should only be used for the intended lagal uses. *(*or any other relevant conclusion *1½ marks)*

Total: 15 marks

10. Intro: Tuberculosis (TB) is a bacterial disease that mainly affects the lungs and causes serious illness *(01 mark)*

Cause: TB is caused by a bacterium called *mycobacterium tuberculois*/ caused by bacteria that multiply in the lungs and cause disease there and the infection may spread to other parts of the body outside the lungs extra-pulmonary tuberculosis) *(01 mark)*

Transmission

Tuberculosis is transmitted through air when a sick person coughs, sneeze, sing, etc., which can put tiny droplets with germ in the air. When another person breathes in the contaminated droplets, the germ enter the lungs and may cause a disease in that person. *(04 marks)*

Signs and symptoms:

- Low fever, tiredness and cough
- Coughing up blood or mucus
- Chest pain, painful breathing or coughing *(Any five 01@ =05 marks)*
- Fever and chills
- Night sweat
- Fatigue and loss of appetite
- Abnormal inflammation of lymph nodes.
- Abnormal mental condition signs.

Control and treatment:

- Isolation
- wearing face mask
- Enough ventilation *(Any four 01@ =04 marks)*
- Covering mouth when coughing and sneezing.
- Vaccinantion.
- TB medications

Conclusion: TB leads to a serious illness. Identifying risk factors may be the best way to avoid TB… (or any other relevant conclusion: *01 mark)* ***Total: 15 marks***

11. Blood transfusion

Blood transfusion is the transfer of blood from a healthy person to a sick person. The one who offer blood for transfusion is known as Donor and the one receiving the blood is known as the Recipient. *(01 mark)*

A blood type (blood group) is a classification of blood based on the presence or absence of inherited antigenic substances on the surface of red blood cells . There is the ABO blood group system and the Rhesus factor (RhD). *(01 mark)*

The blood types in the ABO system are A, B, AB and O (with +, − or Null denoting Rh status) named using the antigens present on the red blood cells. The antigens are associated with anti-A and anti-B antibodies which are found in the plasma. Group O means no antigen on the red blood cells and group AB means no antibodies in the plasma. *(01 ½ marks)*

The Rh system (Rh meaning *Rhesus*) is based on the RhD antigen which is able to provoke an immune system response. Generally, the RhD-negative individuals are not able to provoke the production of anti-D antibodies, while RhD-positive are. The presence or absence of the Rh antigens is signified by the + or − sign, so that for example the A− group does not have any of the Rh antigens. *(01 ½ marks)*

Blood transfusion is very important since it makes possible to save lives of those who lost a lot of blood from accidents and other problems. *(01 mark)*

The blood for transfusion must be compatible, thus blood transfusion involves testing blood from both donors and recipients to ensure that every individual recipient is given blood that is compatible and is as safe as possible. If a unit of incompatible blood is transfused between a donor and recipient, a severe acute hemolytic reaction with hemolysis (RBC destruction), renal failure and shock is likely to occur, and death is a possibility. *(02 marks)*

One way of checking for compatibility is mixing a sample of the recipient's serum with a sample of the donor's red blood cells and checking if the mixture agglutinates, or forms clumps. *(01 mark)*

- Individual with type AB blood can receive blood from any group (with AB being preferable), but cannot donate blood to any group other than AB. They are known as universal recipients.
- Group A individual can receive blood only from individuals of groups A or O, and can donate blood to individuals with type A or AB.
- Group B individual can receive blood only from individuals of groups B or O, and can donate blood to individuals with type B or AB. *(01@ =05 marks)*
- A group 'O' individual can receive blood only from a group 'O' individual, but can donate blood to individuals of any ABO blood group (i.e., A, B, O or AB).
- And by the **Rhesus system**, Rh-positive donor will cause adverse reaction in Rh –negative patient by provoking the formation of anti-Rh antibodies. Rh -positive patients do not react to Rh-negative blood. Thus Rh- can safely donate to a Rh+ while the opposite does not work. This means also that A+ should not be transfused into A-, or AB-.

Conclusion: Blood transfusion is important in saving people's lives, however, precaution regarding compatibility must be observed in order for the transfusion to be safe (or any other relevant conclusion: *01 mark*)

Total: 15 marks